

S_T function under GBM

$$\log\left(\frac{S_T}{S_0}\right) \sim N(mT, \sigma^2 T)$$

from mean and variance
scaling linearly with T (assuming independence of increments)

$$S_T = S_0 e^x \quad \text{with } x \sim N(mT, \sigma^2 T)$$

$$\begin{aligned} E[S_T] &= S_0 E[e^x] \\ &= S_0 e^{mT + \frac{\sigma^2 T}{2}} \\ &= S_0 e^{rT} \end{aligned}$$

$$E[S_T] = S_0 e^{rT}$$

$$\Rightarrow rT = mT + \frac{\sigma^2 T}{2}$$

$$m = r - \frac{\sigma^2}{2}$$

$$S_T = S_0 e^x \quad \text{with } x \sim N\left(\left(r - \frac{\sigma^2}{2}\right)T, \sigma^2 T\right)$$

$$Z = \frac{x - \left(r - \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} \quad \text{with } Z \sim N(0,1)$$

$$S_T = S_0 \exp\left[\left(r - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}Z\right] \quad \square$$